

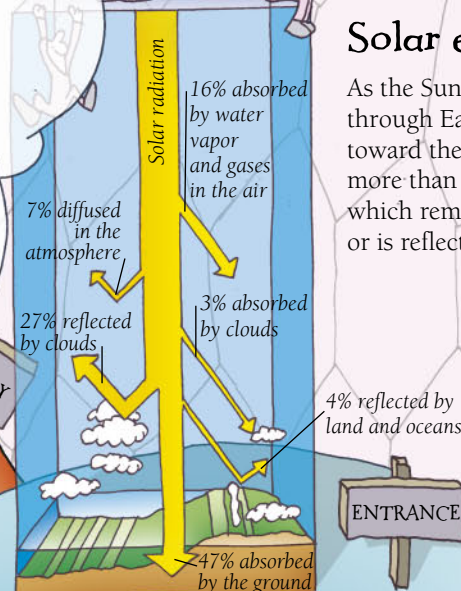
WEATHER & CLIMATE

An area's weather is what happens from day to day. It can be unpredictable and change quickly, whereas climate is defined by the typical weather recorded in an area for a period of 30 years or more. Climate is affected by latitude, height above sea level, the region's prevailing wind, and the circulation of ocean currents that warm or cool the air around them.

I'm full of energy.

Solar energy

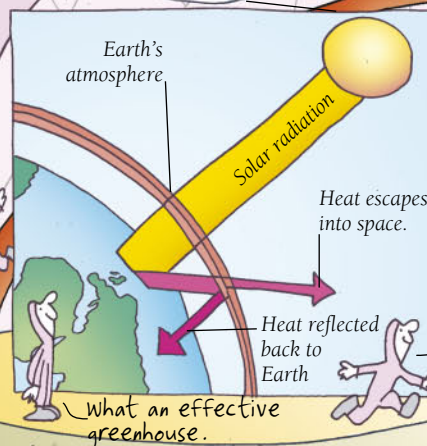
As the Sun's rays beam down through Earth's atmosphere toward the ground, they lose more than half their energy, which remains in the atmosphere or is reflected back into space.



I've got a sunny disposition.

Greenhouse effect

Gases in Earth's atmosphere trap the Sun's heat, just like a greenhouse. Some human activity releases more "greenhouse gases" into the atmosphere, increasing the greenhouse effect and affecting Earth's weather, which could result in climate change.



This way!

Down in front!

Clouds form along the warm front.

WARM FRONT

COLD FRONT

Thunderstorms and rain occur along the cold front.

Everyday weather

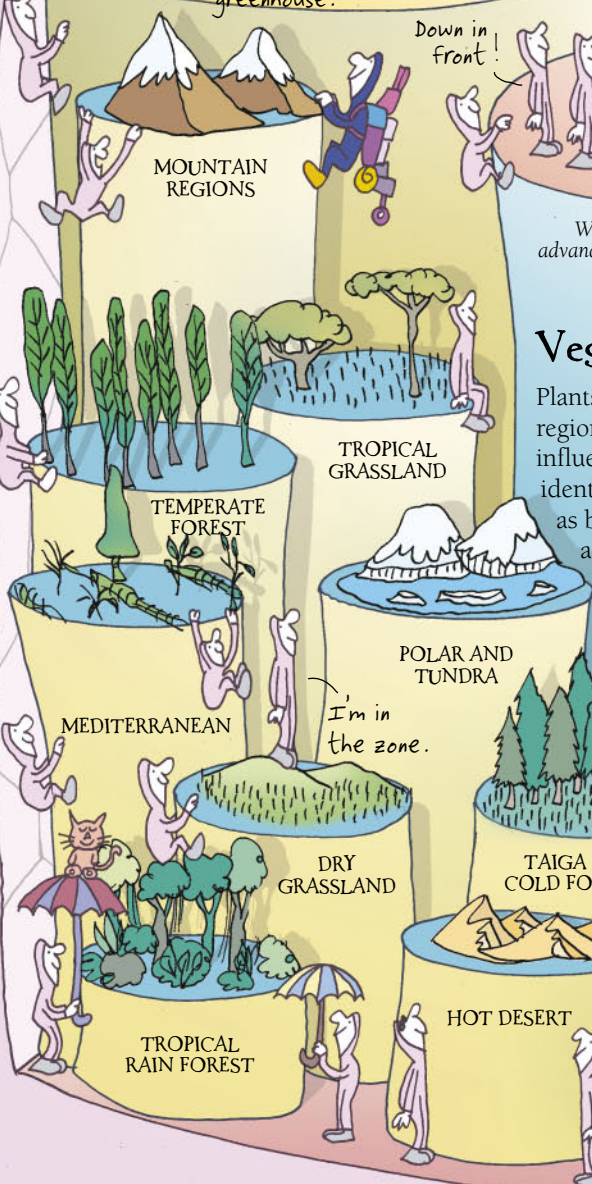
Air masses in the atmosphere affect our day-to-day weather. As they pass overhead, they bring stable, unchanging weather. However, when one air mass collides with another, it causes a change in the weather along the boundary, or front, between them.

I'm way out in front!

THIS WAY

Vegetation zones

Plants and animals vary massively from region to region across Earth, strongly influenced by climate. Scientists have identified nine basic environments known as biomes, each of which has a typical array of plants and animals that have adapted to survive within it.



It's a perfect storm!

Lightning flashes from a negatively charged to a positively charged area.

LOOK UP

Storms

Severe weather, including strong winds, driving rain, and thunder and lightning, is often called a storm, although a storm is scientifically defined as an event where winds reach more than 55mph (88kph)—Force 10 on the Beaufort scale.

Thunder and lightning

A flash of lightning from an electrically charged thundercloud heats the air and causes it to expand, creating a clap of thunder.

Hurricanes

These hugely destructive weather systems are a mass of violent tropical storms that cluster around a ring of low pressure over water.